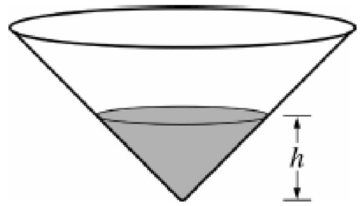
Name ________________________________ **Problem Set 14**

"Do the problem sets honestly. If you cheat and look up the answers, it will not help you whatsoever. Practicing free responses again and again exposes you to a ton of different question possibilities." A 2019 Nerd

FR1. Practice AP Test 2016

A graphing calculator is required for problems on this part of the exam.



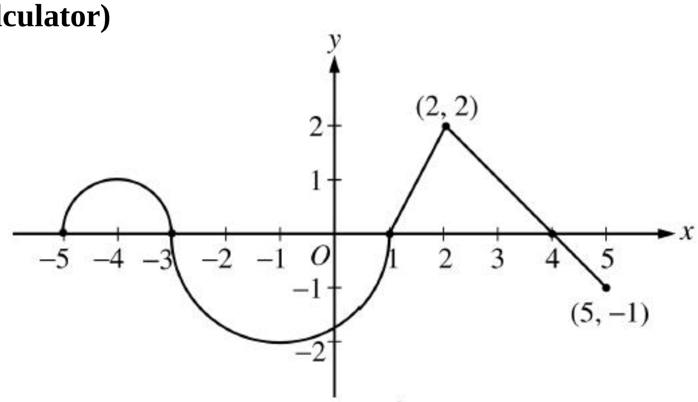
The height of the water in a conical storage tank, shown above, is modeled by a differentiable function h, where $h(t)$ is measured in meters and t is measured in hours. At time $t = 0$, the height of the water in the tank is 25 meters. The height is changing at the rate $h'(t) = 2 - \frac{24e^{-0.025t}}{t+4}$ meters per hour for $0 \le t \le 24$.

(a) When the height of the water in the tank is h meters, the volume of water is $V = \frac{1}{3}\pi h^3$. At what rate is the volume of water changing at time $t = 0$? Indicate units of measure.

(b) What is the minimum height of the water during the time period $0 \le t \le 24$? Justify your answer.

(c) The line tangent to the graph of h at $t = 16$ is used to approximate the height of the water in the tank. Using the tangent line approximation, at what time t does the height of the water return to 25 meters?

FR1. 2007B BC 4 **(No Calculator)**



Graph of f'

Let f be a function defined on the closed interval $-5 \le x \le 5$ with $f(1) = 3$. The graph of f', the derivative of f, consists of two semicircles and two line segments, as shown above.

a) For $-5 < x < 5$, find all values of x at which f has a relative minimum. Justify your answer.
b) For $-5 < x < 5$, find all values of x at which f has a point of inflection. Justify your answer.
c) Find all intervals on which f is concave down and also is increasing. Explain your reasoning.
d) Find the absolute maximum value of f over the closed interval $-5 \le x \le 5$. Explain your reasoning.

MC1. $\lim\limits_{x\to\pi} \dfrac{\cos x+\sin(2x)+1}{x^2-\pi^2}$ is

(A) $\dfrac{1}{2\pi}$

(B) $\dfrac{1}{\pi}$

(C) 1

(D) nonexistent

MC2. $\displaystyle\int_2^3 \frac{3}{(x-1)(x+2)}\,dx =$

(A) $-\dfrac{33}{20}$ (B) $-\dfrac{9}{20}$ (C) $\ln\left(\dfrac{5}{2}\right)$ (D) $\ln\left(\dfrac{8}{5}\right)$

MC3. A curve in the plane is defined parametrically by the equations $x=t^3+t$ and $y=t^4+2t^2$. An equation of the line tangent to the curve at $t=1$ is

(A) $y=2x$ (B) $y=8x$ (C) $y=2x-1$

(D) $y=4x-5$ (E) $y=8x+13$

MC4. Let g be a continuous function. Using the substitution $u = 2x - 1$, the integral $\int_2^3 g(2x-1)\,dx$ is equal to which of the following?

(A) $\int_2^3 g(u)\,du$ (B) $\dfrac{1}{2}\int_2^3 g(u)\,du$ (C) $\int_3^5 g(u)\,du$ (D) $\dfrac{1}{2}\int_3^5 g(u)\,du$